Enhanced BSL Containment Architecture

Tiered Physical Escalation with Human-Failure-Resilient Balance Layers

Version 2.0 – Transmission Vector Integration

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This document is published as a contribution to public discourse on biocontainment infrastructure. It is intended for review by biosecurity professionals, policy makers, and the interested public. All technical specifications are proposed for evaluation and are not validated by operational testing.

The views expressed are those of the author and do not represent any organisation.

Version 2.0 – Public Review Draft. Comments and substantive critiques are invited for incorporation into future versions.

VERSIONING PHILOSOPHY

This document follows a “no updates, only upgrades” methodology. Version 2.0 represents a substantive upgrade from the original architecture — adding detailed transmission vector taxonomy, airborne transmission mechanics, routine infection pathway analysis, treaty architecture prerequisites, tier transition transport protocols, and facility decommissioning procedures. Minor corrections will be applied silently. No version 2.1, 2.2, or similar incremental patches will be issued. Comments and substantive critiques are invited and will be incorporated into the next major version with attribution unless anonymity is requested.

Contents

1 DEFINITIONS

- ***Human dumbassery***
  A formal design hazard encompassing: fatigue-driven procedural shortcuts, single-point human decision failures, institutional face-saving delays, deliberate narrative control over transparent escalation, and the statistically predictable tendency of tired, career-incentivised, or cognitively overloaded humans to mishandle high-stakes containment events.

- ***Transmission vector***
  Any mechanism by which a pathogen moves from a reservoir or infected host to a new host. Transmission vectors are categorised in this document as: airborne (aerosol, droplet nuclei, inspirable particles), droplet (ballistic large particles, typically $> 5\,\mu$m, limited range), fomite (contaminated surfaces and objects), direct contact (blood, bodily fluids, tissue), vector-borne (arthropod intermediate), and environmental (water, soil, air-handling systems, persistent surface deposition).

- ***Airborne transmission***
  Transmission via particles small enough to remain suspended in air for extended periods, travel beyond the immediate source area, and be inhaled into the lower respiratory tract. This includes aerosol transmission (particles typically $< 5\,\mu$m that can remain suspended for minutes to hours and travel significant distances) and droplet nuclei (desiccated residues of larger droplets that shrink to inspirable size). Airborne transmission is the highest-concern category for containment because it cannot be blocked by standard droplet precautions, enables superspreader events, and allows infection at distance from the source.

- ***Routine transmission pathway***
  The mundane, repeatable chain of events by which a pathogen escapes containment and establishes infection in a new host outside the facility. Routine pathways are distinguished from catastrophic pathways (explosion, structural failure, deliberate attack) by their banality: contaminated shoes, shared equipment, incomplete decontamination, fatigued procedural lapse, aerosol generation during routine manipulation, HVAC recirculation, waste handling error. Routine pathways are responsible for the majority of documented laboratory-acquired infections and are the primary target of the balance layers in this architecture.

- ***Fomite half-life***
  The time required for a 50% reduction in viable pathogen on a specified surface under specified environmental conditions (temperature, humidity, surface material). Fomite half-life is a critical escalation trigger in the Tier Qualification Matrix (Section 7).

- ***Superspreader event***
  A single infected individual transmitting to a disproportionate number of secondary cases, typically driven by a combination of high viral load, asymptomatic or pre-symptomatic shedding, airborne transmission in a confined or poorly ventilated space, and high contact density.

- ***Escalation tiers***

A spectrum of physically separated, progressively de-humanised containment environments:

  - BSL-4 Enhanced – current multi-agent facilities with mandatory layered human/technical interlocks
  - BSL-5 Remote Terrestrial – hardened, minimal-human, robotically-operated remote sites
  - BSL-6 Automated Ground – fully robotic underground or bunker installations with zero routine human entry
  - BSL-7 Deep Orbital – distant stable orbit module; autonomous breach-destruction sequence; zero material return to Earth

- ***Balance layers***
  Governance, technical, operational, incentive, and human-factors engineering mechanisms stacked on top of the physical tiers. Their sole purpose is to reduce reliance on perfect human behaviour and to ensure that when a person — inevitably — cuts a corner, suppresses an alert, or delays a decision, the system does not cascade into global seeding.

- ***Dual-AI disagreement threshold***
  A calibrated Kullback-Leibler divergence value representing the maximum acceptable divergence between two independent AI system assessments of facility state. When the divergence exceeds this threshold for more than a specified time window, automatic escalation is triggered.

- ***Dead-man switch***
  An autonomous safety mechanism that executes a pre-authorised action (lockdown, decontamination, sterilisation, destruction) unless a regularly-timed positive confirmation signal is received from the required number of authorised humans and AI instances. Failure to reset within the window triggers the action automatically. The reset window tightens with increasing tier.

2 EXECUTIVE SUMMARY

Current BSL-4 practice co-locates parent agents and routinely publishes chimeric construction techniques capable of combining high lethality, aerosol/respiratory transmission, and environmental stability. Documented mundane escape vectors — contaminated shoes, shared equipment, decon shortcuts, incomplete doffing, HVAC recirculation errors — and real-world early-signal suppression patterns leave the existing system acutely vulnerable to exactly the type of human dumbassery that turns a contained incident into a global event.

The transmission characteristics of greatest concern are well-characterised in the epidemiological literature. Airborne transmission via aerosol particles $< 5\,\mu$m enables infection at distance, superspreader dynamics, and penetration of standard droplet precautions. Extended fomite half-life on common surfaces — stainless steel, plastic, glass, fabric — creates a persistent contamination reservoir that can infect hours or days after the initial deposition event. Routine transmission pathways — the mundane chains of contact, surface transfer, and inhalation that occur during normal laboratory operations — are responsible for the majority of documented laboratory-acquired infections. These pathways do not require dramatic failures. They require only a single fatigued procedural lapse at the wrong moment.

This updated architecture adds the full escalation ladder through BSL-7 Deep Orbital, integrates detailed transmission vector taxonomy and routine pathway analysis, and embeds twelve interdependent balance layers, every one of them designed under the assumption that humans will get tired, protect their institutions, and delay bad news. The system does not ask for better humans. It compensates for them through mandatory dual verification, immutable independent logging, pre-authorised automated escalation, fatigue-based interlocks, and incentive structures that reward early flagging rather than perfect containment optics.

The Tier Qualification Matrix (Section 7) has been expanded to incorporate specific transmission characteristics — airborne particle size distribution, fomite half-life thresholds, superspreader potential, routine pathway exploitability — as mandatory escalation triggers. A construct that combines respiratory transmission with extended environmental stability escalates automatically, regardless of institutional risk assessment.

The conclusion is straightforward: the research capable of creating the worst hybrid constructs is already underway in shared facilities. The transmission pathways those constructs would exploit are well-understood and repeatedly documented. The engineering and governance response must now match the threat by eliminating the single-point human failure modes that have repeatedly shown themselves in lower-consequence events.

3 THREAT LANDSCAPE & FAILURE MODE MAPPING

3.1 Pathogen Toolkit Already Present

Multi-agent BSL-4 facilities currently run Ebola/Marburg filoviruses, Nipah henipavirus, mammalian-adapted H5N1 influenza, SARS-like coronaviruses, Lassa and other arenaviruses, and CCHF nairovirus side-by-side in shared equipment environments. Published reverse-genetics systems (circular polymerase extension cloning, Gibson assembly, yeast-based reconstruction) and pseudotyping methods (VSV, lentiviral, retroviral backbones) allow constructs combining:

- A coronavirus or influenza respiratory backbone conferring efficient aerosol transmission, high viral load in upper respiratory tract during pre-symptomatic phase, and droplet nuclei production during normal speech and breathing
- Filovirus or arenavirus glycoproteins conferring broad tissue tropism, immune evasion, endothelial disruption, and case-fatality rates of 40–90%
- Paramyxovirus or henipavirus fusion proteins conferring syncytia formation, direct cell-to-cell spread bypassing antibody neutralisation, and neurological involvement
- Environmental stability genes from poxviruses or parvoviruses conferring extended fomite survival on stainless steel, plastic, glass, and fabric surfaces

The resulting hybrid transmits efficiently via aerosols, persists for hours to days on common surfaces, sheds during the pre-symptomatic phase, and kills a large fraction of those infected. This is not speculative. The component parts exist in current research collections. The assembly methods are published.

3.2 Transmission Vector Taxonomy

3.2.1 Airborne Transmission — Aerosol and Droplet Nuclei

Particles $< 5\,\mu$m in aerodynamic diameter. Remain suspended in air for minutes to hours depending on particle size, air currents, and ventilation. Travel beyond the immediate source area — documented transmission at distances exceeding 6 metres in indoor environments with poor ventilation. Penetrate standard surgical masks. Require N95/FFP3 or PAPR respiratory protection for safe handling. Deposition occurs throughout the respiratory tract with particles $< 2.5\,\mu$m reaching the alveolar region.

Airborne transmission is the dominant concern for pandemic potential because it enables:

- Superspreader events in confined spaces
- Pre-symptomatic transmission before the index case is identified
- Infection at distance without direct contact
- Healthcare system overload through rapid exponential case growth
- Evasion of standard droplet precautions

3.2.2 Droplet Transmission

Particles $> 5\,\mu$m in aerodynamic diameter. Ballistic trajectory — typically fall to ground or surfaces within 1–2 metres of the source. Generated during coughing, sneezing, and certain medical procedures. Blocked by standard surgical masks. Limited range makes droplet-transmitted pathogens easier to contain through distancing and barrier precautions.

Note: The $5\,\mu$m cutoff between droplet and airborne is a convention, not a physical law. Particles in the 5–20 μm range can remain suspended under certain humidity conditions and air currents.

3.2.3 Fomite Transmission

Transmission via contaminated surfaces and objects. Pathogen deposited on a surface by contact with infected material, then transferred to a new host through hand-to-face contact, contact with mucous membranes, or contact with an open wound. ***Fomite half-life*** is the critical variable.

Documented fomite half-lives of concern:

- Influenza A on stainless steel: 24–48 hours
- SARS-CoV-2 on stainless steel and plastic: 72 hours
- Ebola on glass: 50 days under cool, dark conditions
- Parvovirus on environmental surfaces: months to years

- *Bacillus anthracis* spores: decades

Fomite transmission is the most mundane and therefore most likely escape vector. Contaminated shoes, shared equipment, incomplete decontamination, improper doffing, and waste handling errors have all been documented.

3.2.4 Direct Contact Transmission

Transmission via direct contact with infected blood, bodily fluids, tissues, or laboratory samples. Primary concern during animal handling, necropsy, sample processing, and needlestick injuries. Standard barrier precautions provide effective protection when properly used. Failure modes are procedural.

3.2.5 Vector-Borne Transmission

Transmission via arthropod intermediate — mosquitoes, ticks, fleas, sandflies. Relevant if the construct includes a vector-competent backbone and the facility is located in an area where the vector species is present.

3.2.6 Environmental and Air-Handling Transmission

Transmission via building systems — HVAC recirculation, contaminated water supply, aerosolisation from drains or sink traps, air-handling unit contamination. A single contaminated surface in an air-handling unit can distribute viable pathogen throughout a facility or to the surrounding environment.

3.3 Routine Transmission Pathways — The Mundane Escape

The following routine pathways are responsible for the majority of documented laboratory-acquired infections.

Pathway 1: Contaminated footwear and clothing
Pathogen deposited on floor or work surface. Transferred to shoe covers, gown cuffs, or clothing during routine movement. Exits containment on personnel.

Pathway 2: Incomplete decontamination of shared equipment
Equipment used with one agent, decontaminated, then used with a different agent or removed from containment. Incomplete decontamination leaves viable pathogen in seams.

Pathway 3: Improper doffing sequence
Fatigued or distracted personnel remove PPE in incorrect order, contaminating hands or clothing. The doffing sequence is the highest-risk routine procedure in BSL-4 operations.

Pathway 4: Aerosol generation during routine manipulation
Pipetting, vortexing, centrifugation, sonication, and syringe manipulation generate aerosols. Manipulation outside the biosafety cabinet due to ergonomic difficulty, haste, or fatigue can release aerosols.

Pathway 5: Sharps injury
Needlestick, scalpel laceration, broken glass. Direct inoculation of pathogen into tissue.

Pathway 6: Waste handling error
Incomplete autoclave cycle, improper waste packaging, leak during transport.

Pathway 7: HVAC malfunction or contamination
HEPA filter failure, incorrect pressure cascade, air-handling unit contamination.

Pathway 8: Exit screening failure
Personnel exit containment while unknowingly contaminated or during the incubation period of an infection acquired inside the facility.

3.4 Highest-Probability Real-World Escape Vectors

Vector A: Mundane fomite escape — contaminated shoes, cuffs, or equipment exit containment. Index case infects community before diagnosis.

Vector B: Single-person procedural lapse — fatigued technician bypasses a decon step with no witness.

Vector C: Institutional suppression window — doctor-level signal → internal discussion → institutional delay → late external notification.

Vector D: Pre-symptomatic or asymptomatic shedding — worker infected inside facility, sheds during incubation before symptoms appear.

3.5 Documented Suppression Archetype

Pattern (repeatable and well-documented across multiple national contexts):
Doctor-level first signal → internal research discussion → institutional delay while origin is investigated, legal liability is assessed, and PR statements are prepared → eventual external notification days or weeks after targeted containment could have been effective → public health authorities begin contact tracing long after the index case's contacts have dispersed.

This is not a hypothetical. This is the documented timeline for multiple laboratory-acquired infection events and the early weeks of several recent epidemics. Any new hybrid escape will encounter the same institutional dynamics unless the notification loop is removed from institutional control entirely.

3.6 Trend Lines

Volume of gain-of-function and chimeric work is increasing globally. Published methods for reverse genetics, pseudotyping, and synthetic biology are becoming more accessible. Human-error failure modes remain static. Without added layers of engineered protection, the probability of the exact *Contagion*-style scenario rises monotonically.

4 PHYSICAL ESCALATION TIERS — DETAILED SPECIFICATION

4.1 BSL-4 Enhanced

- **Mandatory dual-human presence** for every high-risk manipulation. Both humans must be present in the suite, visually monitoring each other, with independent authority to halt any procedure.
- **Immutable real-time digital twin logging.** Every action logged sensor-side with cryptographic timestamping. No post-hoc editing possible. Logs streamed to an off-site independent repository.
- **AI camera review.** Computer vision models flag deviations (glove touches face, incorrect doffing, manipulation outside cabinet) and trigger immediate haptic alert + automatic lock of containment doors.
- **Enhanced fomite protocol:** double-layer footwear with inter-layer disinfectant bath; boot scans for fluorescent tracer; full-coverage impermeable gowns with taped cuffs; separated air-handling with real-time airborne particle surveillance.
- **Exit protocol:** mandatory chemical shower with verified contact time; independent verification by AI camera + second human.
- **Reporting obligation:** any deviation must be logged and reported to the oversight body within 4 hours.

4.2 BSL-5 Remote Terrestrial

- **Location:** hardened remote sites at minimum 50 km from populated areas; subterranean where feasible.
- **Personnel footprint:** maximum 3 on-site; robotic primary work via telemanipulation from a separated control room.
- **Permanent quarantine on exit:** personnel exit into a sealed transit habitat for 21-day observation with independent external medical monitoring.
- **Independent international sensor access:** treaty-level obligation for real-time read-only feeds of all environmental sensors to the oversight body.
- **Pre-authorised destruction:** facility-level thermobaric charge array on a 72-hour dead-man timer. Failure to reset triggers automatic full-facility sterilisation.
- **Transport:** dedicated sealed vehicle with onboard environmental monitoring and independent dead-man timer.

4.3 BSL-6 Automated Ground

- **Fully robotic underground or remote bunker.** Zero routine human entry. All manipulations performed by dexterous robots inside a helium-sealed, multi-chamber vault.

- **AI anomaly detection:** multiple independent AI systems (different architectures, different training data) analyse all data streams. Disagreement beyond calibrated KL divergence threshold for >15 min triggers automatic escalation.
- **Human access protocol:** physically impossible for any single individual. Requires live, simultaneous dual-key activation from two separate internationally-appointed oversight representatives in different countries.
- **Construct thresholds:** reserved for any hybrid that combines a respiratory-transmission backbone with high-virulence traits and confirmed fomite half-life exceeding 4 hours on common surfaces.
- **Waste handling:** dual-redundant autoclaves with independent cycle verification; no waste leaves the vault without verified sterilisation.
- **Air handling:** multi-stage HEPA filtration with redundant filter banks; continuous air sampling with PCR analysis of captured particles.

4.4 BSL-7 Deep Orbital — Distant Plane + Active Destruction

4.4.1 Orbital Station Design

- **Location:** Earth-Sun L4 or L5 Lagrange point (∼150 million km from Earth) or distant retrograde lunar orbit (>400,000 km separation).
- **Module composition:** multi-layer vault with independent robotic manipulators; no life-support systems — the station is categorically uninhabitable. No docking port compatible with crewed spacecraft.
- **Operations:** fully robotic, teleoperated from Earth via secure quantum-encrypted link. The approximately 500-second (8.3-minute) light-delay enforces deliberate, pre-planned manipulation sequences.
- **Station-keeping:** autonomous propulsion system; if propellant drops below the destruction manoeuvre requirement, the station executes the destruction sequence pre-emptively.

4.4.2 Destruction Triggers (Autonomous)

- **Ascent-Phase Vehicle Failure Detection:** During launch ascent, the containment vessel monitors redundant sensor feeds from the launch vehicle's telemetry bus. Loss of the heartbeat signal for the pre-defined timeout period (15–30 s) triggers autonomous thermobaric sterilisation of the primary bio-containment volume, followed by controlled structural vaporisation. All fragments must undergo complete atmospheric burn-up. (See Appendix A.)
- **Breach detection:** any combination of unexpected pressure drop, biosensor detection outside the vault boundary, loss of communication heartbeat >60 s, uncommanded station manoeuvre, power system failure exceeding backup capacity, or thermal anomaly in the containment vault.
- **Mission-end or pre-emptive command:** a destruct order can be issued by the international oversight body with dual-human + dual-AI confirmation. National

veto is explicitly disallowed. Any signatory nation may request a destruct order; if the review is not completed within 4 hours, destruction proceeds automatically.

4.4.3 Destruction Sequence

Step 1: Manoeuvre – autonomous propulsion system executes a pre-calculated burn to a clean heliocentric disposal trajectory. Trajectory confirmed by ground-based independent tracking from at least three separate tracking stations.

Step 2: Thermobaric sterilisation – multipoint overpressure charges distribute a uniform 6-bar overpressure wave with peak temperature $> 800\,°\mathrm{C}$ throughout the interior, sustained for minimum 30 minutes.

Step 3: Structural vaporisation – high-energy plasma-arc array or compact laser ablation system transforms remaining structural material into a dispersed atomic gas cloud. No fragment exceeds $100\,\mu$m.

Step 4: Verification – ground-based spectral analysis from multiple independent observatories confirms complete conversion to atomic species.

Step 5: Post-destruction monitoring – dispersed gas cloud tracked for 30 days to confirm trajectory stability.

5 INTEGRATED BALANCE LAYERS

The following twelve balance layers are hardwired architectural constraints, each mapped to a specific documented human failure mode, and each defaults to safety when humans fail to act.

5.1 Technical Balance Layers

5.1.1 Mandatory AI + Human Dual Verification

Every critical decision at BSL-5 and above requires dual independent AI confirmations (separate vendors, different architectures) plus two authorised humans. A single-human override is architecturally impossible.

5.1.2 Immutable Quantum-Sealed Logging

All sensor feeds, camera frames, actuator commands, access events, and decision logs are signed with a NIST-approved post-quantum signature scheme. Signed data is hashed into a continuously published transparency log accessible in real time to the oversight body.

5.1.3 Continuous Fatigue and Cognitive-Load Monitoring

Any humans still involved at BSL-4 Enhanced and BSL-5 are subject to continuous monitoring using multiple independent modalities:

- Eye-tracking based micro-sleep detection (PERCLOS metric)
- EEG headband or fNIRS caps monitoring cognitive load indices

- Keyboard/mouse dynamics and voice analysis
- Heart rate variability monitoring

When a pre-validated fatigue threshold is crossed, the system automatically locks out the individual from critical controls and logs the event immutably. No override is possible.

5.1.4 Dual-AI Anomaly Detection with Automatic Disagreement Escalation

Two independently developed AI systems continuously evaluate the facility state. If their assessments disagree beyond a calibrated Kullback-Leibler divergence threshold for more than 60 seconds, the system triggers an automatic escalation standby.

5.1.5 Dead-Man Switch Architecture

Every tier above BSL-4 incorporates a tier-appropriate destruction or decontamination sequence that executes automatically if a regularly-timed positive confirmation signal is not received:

- BSL-5 Remote: 72-hour reset window, dual-human + dual-AI confirmation required.
- BSL-6 Automated: 24-hour reset window, dual-human + dual-AI confirmation required.
- BSL-7 Deep Orbital: 60-second communications heartbeat. If lost, autonomous destruction initiates.

5.2 Governance and Treaty Balance Layers

5.2.1 Independent International Biocontainment Oversight Body (IIBOB)

The IIBOB is composed of rotating national experts, permanent technical staff with no national affiliation, and designated ethicists. It has real-time read-only access to all sensors, logs, camera feeds, and decision records for BSL-5 and above. No national veto on anomaly reporting exists. Funding is automatic and not subject to annual re-approval.

5.2.2 Automatic Trigger Protocols (Pre-Ratified)

Any confirmed breach or unexplained anomaly at BSL-5+ automatically initiates the next tier's response sequence unless overridden by dual independent human + dual AI confirmation within a strict time window:

- BSL-5: 12-hour override window
- BSL-6: 2-hour override window
- BSL-7: Zero override — destruction sequence is irreversible once initiated

5.2.3 Strengthened Whistleblower Protections

Any person who flags a containment concern receives automatic career and legal immunity. Identity is cryptographically sealed and released only to the IIBOB. Penalties for documented suppression include mandatory prosecution for criminal negligence. Positive incentives for early flagging include guaranteed funding prioritisation, co-authorship on safety papers, and formal career protection letters.

5.2.4 Mandatory Near-Miss and Anomaly Disclosure

Every incident above a defined severity threshold must be reported in a redacted public form within 14 days. Opacity itself becomes an automatic IIBOB investigation trigger. If a facility fails to report, the IIBOB issues a global health advisory based on available sensor data.

5.2.5 Treaty Architecture Prerequisites

The IIBOB requires legal instruments that do not currently exist in ratified form. The following are minimum prerequisites before any BSL-5+ facility can be commissioned under this framework:

1. A new Biological Containment Convention (BCC) or equivalent amendment to the Biological Weapons Convention establishing IIBOB with binding authority.
2. Host nation waiver of sovereign veto over IIBOB anomaly reporting.
3. Mandatory jurisdiction: any nation conducting gain-of-function or chimeric research above the thresholds in Section 7 must accede to the BCC.
4. Whistleblower protection treaty.
5. Enforcement mechanism with graduated sanctions.
6. Funding mechanism with mandatory assessed contributions.

Until these instruments are ratified, the governance layers are aspirational. The technical layers function independently and should be implemented regardless of treaty status.

5.3 Operational Resilience Layers

5.3.1 Zero Single-Point Human Decisions

Every action that could lead to containment breach at BSL-5+ requires dual independent confirmation enforced through hardware key-cards that require simultaneous, spatially-separated activations.

5.3.2 Assume-Failure Design Philosophy

Every tier's procedures start from the question: "What happens when the human fails to spot the problem?" All responses are pre-wired to sensor thresholds, not to after-the-fact human diagnosis. Human confirmation can delay but not permanently block an automated safety sequence.

5.3.3 Mandatory Rotation and Rest Protocols

No individual may work high-containment tasks for more than 4 consecutive hours. A minimum 12-hour off-duty period is required before re-entry. Biometric verification enforces this.

5.3.4 Regular Independent Red-Team Exercises

Conducted quarterly by a separate agency. Red teams are tasked with exploiting human shortcut behaviour, social engineering, incentive structures that encourage suppression, and physical security gaps. All findings are published in sanitised form within 30 days and trigger binding procedure updates within 90 days.

5.4 Incentive Redesign Layers

5.4.1 Metrics Shift

Institutional scorecard metrics shift from "days since last incident" to:

- Median time from first internal signal to external notification
- Number of near-miss flags raised and resolved (higher is better)
- Red-team exercise performance
- Fatigue-lockout event rate (increasing rate triggers workload review)
- Whistleblower reports received and substantiated (higher is better)

5.4.2 Automatic External Investigation Trigger

Any delay exceeding 48 hours between the first digitally-logged internal signal of an anomaly and notification of public health authorities triggers an IIBOB-led investigation.

5.4.3 Positive Incentives for Early Flagging

Teams that flag and escalate risks early receive guaranteed funding prioritisation, co-authorship on published safety improvement papers, formal career protection letters, and public recognition.

5.5 Human-Factors Engineering Layer — Mapped Directly to Dumbassery Modes

Human dumbassery mode	Engineered countermeasure	Section
Fatigue-driven procedural shortcuts	Continuous fatigue monitoring → automatic lockout, mandatory handoff	5.1.3, 5.3.3
Single-person procedural failure	Dual-human + dual-AI verification; no single override possible	5.1.1, 5.3.1
Institutional face-saving delay	IIBOB real-time sensor access; automatic public notification triggers	5.2.1, 5.2.2
Narrative control / PR framing before action	Pre-authorised destruction sequences that trigger on sensor thresholds	5.1.5, 5.3.2
Tired humans ignoring subtle warnings	Dual-AI disagreement detection that escalates without human petition	5.1.4

Incentivised suppression of near misses	Positive incentives for early flagging + automatic investigation of lag	5.4.2, 5.4.3
Over-confidence in "we can handle it"	Assume-failure design — all tiers have pre-planned worst-case responses	5.3.2
Ergonomic corner-cutting due to cumbersome PPE	Continuous AI camera review flags protocol deviations; PPE validated for task	4.1, 5.1.3
Groupthink and hierarchy effects suppressing dissent	Dual-key architecture requires spatially separated, independent confirmations	5.1.1, 5.3.1
Complacency after long incident-free periods	Quarterly red-team exercises targeting complacency; automatic procedure updates	5.3.4
Inadequate training masked by infrequent testing	Mandatory simulation-based competency assessment every 90 days	5.3.4
Shift-change communication gaps	Immutable digital twin logging provides continuous, queryable record	4.1, 5.1.2

6 AUTOMATIC ESCALATION AND DESTRUCTION TRIGGER MATRIX

Tier	**Trigger event**	**Automatic action**	**Human override window**	**Override requirement**
BSL-4 Enhanced	Unresolved AI deviation flag >10 min	Suite lockdown, external review notification, automatic air sampling	30 min	Dual-human confirmation (in-suite + external)
BSL-4 Enhanced	Biosensor detection of target nucleic acid outside primary containment	Suite lockdown, automatic notification to national health authority, personnel quarantine in place	2 h	Dual-human + single AI confirmation
BSL-4 Enhanced	Fatigue lockout of sole authorised operator with no rested replacement	Suite lockdown, external review notification, mandatory procedure suspension	None	N/A
BSL-5 Remote	Breach signature confirmed by 1 sensor array + AI	Pre-alert IIBOB; stand-by thermobaric charge active; automatic air and surface sampling	12 h	Dual-human + dual-AI confirmation
BSL-5 Remote	Dual-AI disagreement >60 s	Escalation to external review; facility enters safe mode	4 h	Dual-human + dual-AI from oversight body
BSL-5 Remote	Dead-man timer expiry (72 h without positive reset)	Full thermobaric facility sterilisation	None	N/A

BSL-5 Remote	Communications blackout >6 h	Pre-authorised transition to autonomous safe mode; standby sterilisation charge active	24 h from comms restoration	Dual-human + dual-AI + IIBOB authorisation
BSL-6 Automated	Dual-AI disagreement >15 min	Autonomous full-facility inert gas purge + seal; external review mandatory	2 h	Dual-human + dual-AI (internationally separate)
BSL-6 Automated	Biosensor breach in vault	Vault seal; automatic sterilisation charge standby; IIBOB immediate notification	1 h	Dual-human + dual-AI + IIBOB authorisation
BSL-6 Automated	Dead-man timer expiry (24 h without positive reset)	Vault-wide inert gas purge + seal + sterilisation sequence	None	N/A
BSL-7 Deep Orbital	Any breach signature	Full autonomous manoeuvre + thermobaric + vaporisation sequence	None	N/A
BSL-7 Deep Orbital	Communications heartbeat loss >60 s	Full autonomous manoeuvre + thermobaric + vaporisation sequence	None	N/A
BSL-7 Deep Orbital	Power system failure exceeding backup capacity	Full autonomous manoeuvre + thermobaric + vaporisation sequence	None	N/A

7 TIER QUALIFICATION MATRIX — WHAT TRIGGERS MANDATORY ESCALATION

The following matrix defines mandatory minimum containment tiers based on construct characteristics. Institutional risk assessment may escalate a construct beyond the minimum; it may not downgrade.

7.1 Transmission-Based Criteria

Construct transmission profile	Minimum tier
Wild-type BSL-4 agent handled in accordance with existing protocols; no genetic modification; no respiratory transmission enhancement	BSL-4 Enhanced
Any construct combining genetic material from two or more BSL-4 agents	BSL-5 Remote
Respiratory or aerosol transmission backbone combined with any BSL-4 agent glycoprotein or virulence factor	BSL-5 Remote

Confirmed airborne transmission with particle size distribution showing $> 10\%$ of shed particles $< 5\,\mu$m aerodynamic diameter	BSL-5 Remote min.; BSL-6 Automated if combined with extended fomite stability
Airborne transmission with documented superspreader dynamics ($k > 0.5$ in animal transmission models)	BSL-6 Automated
Airborne transmission with pre-symptomatic shedding confirmed in animal model	BSL-6 Automated
Airborne transmission with asymptomatic shedding confirmed in any model	BSL-7 Deep Orbital

7.2 Fomite Stability Criteria

Construct fomite characteristics	**Minimum tier**
Fomite half-life < 2 h on stainless steel under standard lab conditions (22 °C, 40% RH)	BSL-4 Enhanced
Fomite half-life 2–4 h on stainless steel	BSL-5 Remote
Fomite half-life 4–24 h on stainless steel	BSL-6 Automated
Fomite half-life > 24 h on stainless steel OR > 4 h on porous surfaces	BSL-6 Automated min.; BSL-7 Deep Orbital if combined with airborne transmission
Fomite half-life > 7 days on any common surface	BSL-7 Deep Orbital
Environmental stability genes from poxvirus, parvovirus, or spore-forming bacteria intentionally introduced	BSL-7 Deep Orbital

7.3 Lethality and Pathogenicity Criteria

Construct lethality profile	**Minimum tier**
Case-fatality rate $< 10\%$ in animal model	BSL-4 Enhanced
Case-fatality rate 10–30% in animal model	BSL-5 Remote
Case-fatality rate 30–60% in animal model	BSL-5 Remote min.; BSL-6 Automated if combined with respiratory transmission
Case-fatality rate $> 60\%$ in animal model	BSL-6 Automated min.; BSL-7 Deep Orbital if combined with any respiratory transmission

Neurotropism or neuroinvasion confirmed in animal model	BSL-6 Automated (regardless of CFR)
Immune evasion confirmed — construct escapes neutralising antibody response to parent strain	BSL-6 Automated min.; BSL-7 Deep Orbital if combined with respiratory transmission
Direct cell-to-cell spread (syncytia formation) bypassing humoral immunity	BSL-6 Automated

7.4 Combined Criteria — Automatic BSL-7 Mandate

The following combinations trigger automatic BSL-7 Deep Orbital requirement with no lower tier permissible:

- Airborne transmission (particles $< 5\,\mu$m, superspreader dynamics) + case-fatality rate $> 30\%$ in animal model + fomite half-life > 4 h
- Airborne transmission + pre-symptomatic shedding + case-fatality rate $> 10\%$ + documented environmental stability > 24 h
- Any construct with airborne transmission, case-fatality $> 60\%$, and confirmed immune evasion
- Any construct specifically designed or predicted to achieve pandemic potential in humans
- Any construct where the research protocol explicitly states the goal of creating a transmissible, highly lethal, environmentally stable hybrid
- Any construct for which no existing countermeasure is effective, combined with respiratory transmission

7.5 Research Intent Criteria

Research specifically aiming to create transmissible, highly lethal, stable hybrids — regardless of whether the resulting construct meets the above thresholds — is restricted to BSL-7 Deep Orbital only. No terrestrial tier is permissible.

8 TIER TRANSITION TRANSPORT PROTOCOL

When a construct qualifies for a higher tier than the facility in which it was created or is currently housed, the following transport protocols apply.

8.1 Transport Container Specifications

All transport containers must be:

- Triple-redundant sealed with inter-layer disinfectant reservoirs between each seal layer

- Rated for maximum credible accident scenarios (impact, fire, water immersion, pressure loss)
- Equipped with continuous environmental monitoring and autonomous sterilisation charge on a dead-man timer
- Tracked in real time with data streamed to the IIBOB and the receiving facility

8.2 Transport by Tier

BSL-4 to BSL-5: Ground or air transport in a dedicated convoy with armed escort; geofenced real-time GPS tracking; dead-man timer with 4-hour reset window; route must avoid population centres.

BSL-5 to BSL-6: All previous requirements plus military-grade escort, no-fly-zone declaration, pre-positioned emergency response teams at 50 km intervals; dead-man timer reset window reduced to 2 hours.

Transport to BSL-7 Orbital: Construct transferred to a dedicated orbital launch vehicle at a remote launch site (no population within 100 km of launch trajectory ground track). Payload fairing includes autonomous destruction charge on a 15-minute dead-man timer. Launch trajectory verified by independent tracking from at least three tracking stations. Any deviation triggers autonomous payload destruction.

8.3 Transport Chain of Custody

At every transfer point, dual-human + dual-AI verification confirms container integrity, chain of custody documentation, and receiving facility readiness. A gap in the chain of custody at any point triggers automatic transport abort or emergency sterilisation.

9 FACILITY DECOMMISSIONING PROCEDURES

Each tier requires a decommissioning plan filed with the IIBOB before construction begins. The plan is updated every 5 years.

9.1 BSL-4 Enhanced Decommissioning

Standard decontamination, verification, and decommissioning per existing national guidelines, plus:

- Full chemical decontamination with verified contact time
- Air and surface sampling for target nucleic acid after decontamination
- Independent verification by external biosafety officer
- Decommissioning report filed with IIBOB

9.2 BSL-5 Remote Decommissioning

Full thermobaric sterilisation as a planned procedure:

- Personnel evacuated; facility sealed
- Thermobaric charges detonated; sustained temperature > 800 °C for minimum 60 minutes
- Facility remains sealed for 5 years with remote sensor monitoring
- If no anomaly detected, site may be released; otherwise, perpetual monitoring

9.3 BSL-6 Automated Decommissioning

Full thermobaric or chemical sterilisation of all vault chambers. After sterilisation:

- Facility sealed with backfill material (bentonite clay or equivalent)
- Perpetual monitoring via buried sensors with satellite uplink
- Site permanently restricted — no human access, no repurposing, no excavation
- Restriction legally binding in perpetuity

9.4 BSL-7 Deep Orbital Decommissioning

Identical to the breach-destruction sequence (Section 4.4.3). At mission end or irrecoverable system failure:

- Autonomous manoeuvre to verified heliocentric disposal trajectory
- Thermobaric sterilisation of all internal volumes
- Structural vaporisation to atomic gas cloud
- Spectral verification from ground-based observatories; zero Earth return geometrically guaranteed

No BSL-7 station is allowed to reach a state where it cannot execute its own destruction sequence. If propellant drops to the minimum required for destruction, the station executes decommissioning pre-emptively.

10 COST-RESILIENCE TRADE-OFF

Each tier increases capital cost by roughly an order of magnitude:

- BSL-4 Enhanced: tens of millions USD (retrofit existing facilities)
- BSL-5 Remote: hundreds of millions USD (new construction, remote site)
- BSL-6 Automated: low billions USD (robotic vault, underground construction)
- BSL-7 Deep Orbital: tens of billions USD (orbital launch, Lagrange point station)

The expected cost of a single hybrid escape — measured in mortality, economic collapse, healthcare system destruction, supply chain disruption, and societal trust dissolution — dwarfs the construction and operation of the full ladder. COVID-19 cost the global economy an estimated $12–16 trillion. A hybrid with COVID-19's transmissibility and a

40% case-fatality rate would trigger a cascading collapse that makes economic calculation meaningless.

The BSL-7 architecture is the only configuration that places a literal astronomical barrier between the hybrid and the human population. For constructs that qualify for BSL-7 (Section 7), the cost of the orbital station is an insurance premium against extinction risk.

The balance layers impose operational friction — dual verification slows workflows, fatigue monitoring limits shift flexibility, red-team exercises consume resources. That friction is not a bug. It is the point. A system that operates without friction has not accounted for human dumbassery.

11 IMPLEMENTATION ROADMAP

Phase 1 (0–12 months): Mandate BSL-4 Enhanced protocols; establish IIBOB with treaty backing; commission red-team exercises; publish Tier Qualification Matrix; begin public consultation on BCC.

Phase 2 (12–36 months): Design and site selection for first BSL-5 remote site; retrofit one existing BSL-4 as pilot for dual-AI logging and fatigue monitoring; validate dexterous robotic telemanipulation for BSL-6 workflow; commission BSL-7 mission study; ratify BCC.

Phase 3 (3–7 years): Operationalise first BSL-5 remote site; move all hybrid work above thresholds out of multi-agent urban BSL-4 facilities; complete BSL-6 automated vault design and begin construction; complete BSL-7 orbital station design and contract launch vehicle.

Phase 4 (7–12 years): Build and launch BSL-7 Deep Orbital module; commission BSL-6 automated bunker; achieve full operational capability; conduct first end-to-end drill.

Phase 5 (Ongoing): Quarterly red-team exercises; annual treaty review; continuous sensor monitoring; 5-year decommissioning plan updates; technology refresh.

12 LIMITATIONS, FEASIBILITY, AND ALTERNATIVE MITIGATIONS

The architecture presented in this document contains both immediately actionable elements and elements that remain beyond current technical and political feasibility. This section explicitly delineates those boundaries so that the proposal is not misread as a near-term construction blueprint.

12.1 BSL-7 Deep Orbital as Critical Boundary Condition

The BSL-7 Deep Orbital tier is presented as a thought experiment that defines the absolute outer limit of acceptable risk rather than as an immediately constructible system. Current limitations include:

- Light-speed communication delay of approximately 500 seconds (8.3 minutes) to Earth-Sun L4/L5 Lagrange points renders real-time robotic intervention impossible. All operations must therefore be reduced to pre-planned sequences with no capacity for in-process correction of unexpected events.
- Launch-phase failure protocols requiring internal thermobaric sterilisation followed by structural vaporisation while the payload remains inside a launch vehicle fairing introduce significant debris and safety hazards that have not been resolved by existing launch abort architectures.
- Capital cost on the order of tens of billions of USD for a single uninhabitable, fully autonomous station exceeds any current national or international biosecurity budget allocation.

These constraints do not invalidate the conceptual value of BSL-7. The exercise of designing an architecture that places an astronomical barrier between the highest-risk constructs and the human population forces clarity about the true magnitude of the risk. BSL-7 therefore functions as a boundary condition: any construct that cannot be safely contained under the lower tiers must either be prohibited or accepted only under conditions that currently do not exist.

12.2 Governance Layers as Political Horizon

The Independent International Biocontainment Oversight Body (IIBOB) and the proposed Biological Containment Convention presuppose a level of sovereign cooperation and treaty ratification that does not currently exist. No major power has demonstrated willingness to permanently waive sovereign veto over domestic laboratory operations or to accept binding external authority with enforcement sanctions. These governance elements are therefore aspirational and should be treated as long-term political objectives rather than prerequisites for technical implementation.

The technical balance layers (dual verification, immutable logging, fatigue monitoring, dead-man switches, and automatic escalation triggers) can and should be implemented independently of treaty ratification. They do not require new international law to function.

12.3 Comparison with Alternative Mitigation Investments

An equivalent investment of tens of billions of USD could alternatively be directed toward:

- Accelerated development and global deployment of rapid-response mRNA and other platform vaccine technologies with pre-authorised strain-updating mechanisms.
- Real-time global genomic surveillance networks with mandatory immediate reporting of novel high-risk constructs.
- Strategic national and international stockpiles of broad-spectrum antivirals and monoclonal antibodies.
- Hardened domestic BSL-5 Remote and BSL-6 Automated facilities as described in Sections 4.2 and 4.3.

These alternatives do not eliminate the requirement for improved physical containment of the highest-risk research. They do, however, represent parallel lines of defence that may prove more robust, faster to deploy, and politically achievable than a single orbital station. The architecture in this document does not claim superiority of BSL-7 over such investments; it identifies BSL-7 as the only configuration that geometrically guarantees zero Earth return for constructs that exceed all lower-tier thresholds.

12.4 Recommendation on Use of This Document

This document is best read as a diagnostic and red-team instrument rather than as a construction specification. The Tier Qualification Matrix (Section 7), the mapping of human dumbassery modes to engineered countermeasures (Section 5.5), and the explicit identification of the institutional suppression window as the critical failure point (Section 3.5) are immediately usable by existing BSL-4 facilities and national regulatory bodies. The BSL-5 Remote and BSL-6 Automated concepts represent technically feasible next steps within a 10–15 year horizon. BSL-7 remains a boundary condition that defines when current and near-future containment approaches become fundamentally insufficient.

13 CRITICAL FINDINGS (RESTATED)

Finding 1: Current BSL-4 infrastructure and published hybrid techniques already enable constructs combining airborne transmission, high lethality, and extended environmental stability — exactly those that can exploit mundane escape vectors and early-suppression windows.

Finding 2: The transmission pathways those constructs would exploit are well-characterised and repeatedly documented. They are the known failure modes of existing containment.

Finding 3: The institutional suppression window — the delay between first internal signal and external notification — is the critical variable that determines whether a containment failure becomes a global catastrophe.

Finding 4: The full escalation through BSL-7 Deep Orbital, combined with the twelve-layer balance system, directly targets both the technical hybrid risk and every observed pattern of human dumbassery.

Finding 5: Robustness is achieved by assuming failure at every level and building automatic, pre-authorised, time-bound responses that operate independently of tired, career-conscious, or face-saving decision-makers.

Finding 6: The governance layers require treaty instruments that do not yet exist. The technical layers function independently and should be implemented regardless.

Finding 7: The question is not whether this system is expensive. It is whether doing nothing beyond current BSL-4 while hybrid capability accelerates is a gamble anyone should be willing to take on behalf of the entire species.

Finding 8: The transmission vector analysis is not academic taxonomy. It is the biological substrate on which the engineering architecture rests.

KEY SOURCES

The airborne, fomite, and routine pathway criteria in the Tier Qualification Matrix and the automatic escalation triggers are anchored to the following established sources:

CDC. Scientific Brief: SARS-CoV-2 and Potential Airborne Transmission (updated May 2021). https://www.cdc.gov/coronavirus/2019-ncov/science/science-briefs/sars-cov-2-transmission.html

WHO. Transmission of SARS-CoV-2: Implications for Infection Prevention Precautions (Scientific Brief, 9 July 2020). https://www.who.int/news-room/commentaries/detail/transmission-of-sars-cov-2-implications-for-infection-prevention-precautions

van Doremalen et al. (2020). Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. *NEJM* 382:1564–1567.

Sagripanti et al. (2010). Persistence in darkness of virulent alphaviruses, Ebola virus, and Lassa virus deposited on solid surfaces. *Archives of Virology* 155(12):2035–2039.

Sewell DL (1995). Laboratory-acquired infections. *Clinical Microbiology Reviews* 8(2):357–360.

WHO. Summary of Probable SARS Cases with Onset of Illness from 1 November 2003 to 31 July 2004. https://www.who.int/csr/sars/country/table2004_04_21/en/

He X et al. (2020). Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nature Medicine* 26:672–675.

Lloyd-Smith JO et al. (2005). Superspreading and the effect of individual variation on disease emergence. *Ecology Letters* 8(11):1161–1170.

A APPENDIX A: ASCENT-PHASE FAILURE MODE PROTOCOL — BSL-7 CONTAINMENT VESSEL

A.1 Purpose

This appendix defines the autonomous failure response protocol for the BSL-7 containment vessel during the launch ascent phase (T-0 through orbital insertion). The protocol ensures that any catastrophic failure of the launch vehicle results in immediate, irreversible neutralization of the contained biological material while still at sufficient altitude for complete atmospheric burn-up.

A.2 System Overview

The containment vessel maintains continuous monitoring of redundant sensor feeds from the launch vehicle's telemetry bus. These sensors transmit a persistent heartbeat signal confirming nominal vehicle structural integrity and flight dynamics. The vessel operates under a dead-man switch architecture.

A.3 Dead-Man Switch Logic

As long as the containment vessel continues to receive valid heartbeat signals from the launch vehicle's sensors, it remains in standby. Loss of the heartbeat signal for a pre-defined, non-extendable timeout period (typically 15–30 seconds, calibrated during integration testing) is treated as a catastrophic vehicle failure. The vessel then autonomously triggers its response sequence with no possibility of external override.

A.4 Response Sequence

1. **Internal Thermobaric Sterilization** — Activate internal thermobaric charges to achieve complete neutralization of all biological material inside the primary containment volume.
2. **Structural Vaporization / Controlled Breach** — Once internal sterilization is verified by onboard sensors, the vessel executes controlled vaporization or breach of the outer structure so that remaining fragments are small enough to fully burn up in the atmosphere.
3. **Altitude Assurance** — The sequence is designed to complete at altitude, ensuring no viable material reaches the surface.

A.5 Design Requirements

- The containment vessel must incorporate independent, ruggedized power and sensor systems capable of surviving initial vehicle breakup forces.
- The dead-man switch timeout must be tuned to avoid false triggers during normal flight events while still allowing rapid response.
- Once triggered, the sequence is irreversible.
- All execution data shall be immutably logged for post-incident analysis.

A.6 Implementation Notes

This protocol adds a critical layer of protection during the most vulnerable non-orbital phase of the BSL-7 pathway. It assumes the launch vehicle can fail at any point during ascent and ensures the containment vessel neutralizes itself autonomously.

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Correspondence is welcome from biosecurity professionals, policy makers, researchers, and members of the public.
Please include "BSL Architecture" in the subject line.

APPENDIX B: VIRAL-VECTOR TIER SELECTION DECISION TABLE

This appendix is a conceptual governance aid for Section 7. It does not replace nationally recognised biosafety standards, institutional biosafety committee review, a qualified biosafety officer, or applicable law. The recognised containment requirement for the parent agent, vector system, and procedure is always the minimum floor.

> ***Use rule:*** *determine the provisional tier by the highest triggered row. Two or more concurrent escalation factors, or an unresolved critical uncertainty, require independent review and normally one additional tier of conservatism. Institutional review may escalate the result; it may not downgrade it.*

B.1 Decision sequence

1. Establish the recognised baseline for the parent agent, vector system, procedure, scale, and host.

2. Evaluate every factor in Table B-1; do not average risks across rows.

3. Select the highest provisional tier triggered by any row.

4. Add an uncertainty adjustment when key characteristics are unknown, disputed, or supported only by weak evidence.

5. Require an independent biosafety and legal review before work begins or changes tier.

6. A Tier 7 boundary result means stop or prohibit the work unless a validated, legally authorised facility and governance framework actually exist. If they do not exist, the work does not proceed.

Table B-1. Viral-vector decision aid

Decision factor	Lower-concern profile	Escalation logic and provisional result within this proposal
Recognised baseline	Well-characterised vector and procedure governed by an existing national or institutional standard.	**Floor:** use the recognised requirement. The proposed higher tiers apply only when added characteristics below trigger escalation.
Replication status	Replication-defective system; rescue pathway addressed; controls and inactivation are validated.	**Tier 4 Enhanced:** lower-concern profile within the report's scope. **Tier 5 Remote:** conditional replication or plausible helper rescue. **Tier 6 Automated:** replication-competent or poorly bounded control. **Tier 7 boundary:** replication combined with efficient human spread, severe outcome, and no reliable countermeasure.
Host range and tropism	Narrow, stable host range with no intended increase in human or mammalian tropism.	**Tier 5:** expanded but bounded host range. **Tier 6:** broad, uncertain, or newly human-relevant tropism. **Tier 7 boundary:** deliberate broadening intended or predicted to support efficient human transmission with severe disease.
Transmission and shedding	No credible respiratory transmission; no meaningful shedding expected; exposure opportunities are limited and controlled.	**Tier 5:** aerosol-generating procedure or unresolved shedding question. **Tier 6:** credible respiratory spread, pre-symptomatic shedding, or asymptomatic shedding. **Tier 7 boundary:** efficient respiratory spread combined with severe outcome, persistence, or absent countermeasures.
Payload or resulting phenotype	Reporter, therapeutic, or other payload with no expected increase in virulence, immune escape, or systemic dissemination.	**Tier 5:** bounded immune modulation or cytotoxic effect. **Tier 6:** severe virulence, neurotropism, immune escape, or systemic dissemination. **Tier 7 boundary:** stacked severe traits intended or predicted to create pandemic-scale consequences.
Environmental persistence	Rapidly inactivated under validated conditions; limited environmental persistence and well-defined waste controls.	**Tier 5:** incomplete inactivation evidence or moderate persistence. **Tier 6:** prolonged persistence, prolonged shedding, or unresolved waste pathway. **Tier 7 boundary:** persistence combined with efficient respiratory spread and severe outcome.
Countermeasures	Effective vaccine, antiviral, post-exposure measure, and validated decontamination are available for the relevant hazard.	**Tier 5:** partial or access-limited countermeasure. **Tier 6:** absent, unvalidated, or materially uncertain countermeasure. **Tier 7 boundary:** no effective countermeasure combined with efficient human spread and severe disease.

Decision factor	Lower-concern profile	Escalation logic and provisional result within this proposal
Procedure, scale, and exposure opportunity	Small-scale work in closed systems with validated engineering controls and limited personnel interaction.	**Tier 5:** animal work, aerosol-generating manipulation, or scale-up that materially increases exposure opportunities. **Tier 6:** high-throughput or repeated operations where a single containment lapse could seed wider exposure. **Tier 7 boundary:** no feasible terrestrial workflow can bound the consequence.
Data quality and uncertainty	Critical characteristics are independently characterised with reproducible evidence.	**No downgrade for uncertainty.** One material gap normally escalates review. Conflicting evidence or unknown critical traits normally require **Tier 6 review or a pause**. When risk cannot be bounded and no safe lower-tier path exists, classify as **Tier 7 boundary / do not proceed**.
Research objective	Objective does not seek increased transmissibility, severity, environmental persistence, immune escape, or broad human host range.	**Tier 6 minimum:** the stated objective materially increases one major hazard dimension. **Tier 7 boundary / prohibition:** the objective explicitly seeks a highly transmissible, severe, persistent, or countermeasure-resistant human pathogen phenotype.

B.2 Decision record

Every tier-selection decision should be preserved as an auditable record rather than as an informal judgement. Record the strongest trigger, the evidence supporting it, any uncertainty adjustment, and the independent reviewers who approved the outcome.

Field	Required entry
Vector / construct identifier	Stable internal identifier; avoid relying only on a working filename.
Recognised baseline	Applicable national, institutional, and parent-agent/vector requirement.
Highest triggered factor	The row in Table B-1 that produced the highest provisional tier.
Evidence status	Verified, reported, model-derived, inferred, conflicting, or unknown.
Uncertainty adjustment	None, one-tier escalation, hold pending evidence, or stop/prohibit.
Provisional tier	Tier 4 Enhanced, Tier 5 Remote, Tier 6 Automated, or Tier 7 boundary.
Independent review	Biosafety officer, institutional biosafety committee, legal/regulatory review, and date.
Final disposition	Approved, approved with conditions, deferred, transferred to a higher tier, or prohibited.

Interpretation note: *Tier 7 remains a conceptual boundary condition in this draft. A Tier 7 outcome is therefore primarily a decision not to proceed under existing conditions, not permission to improvise an unavailable facility.*